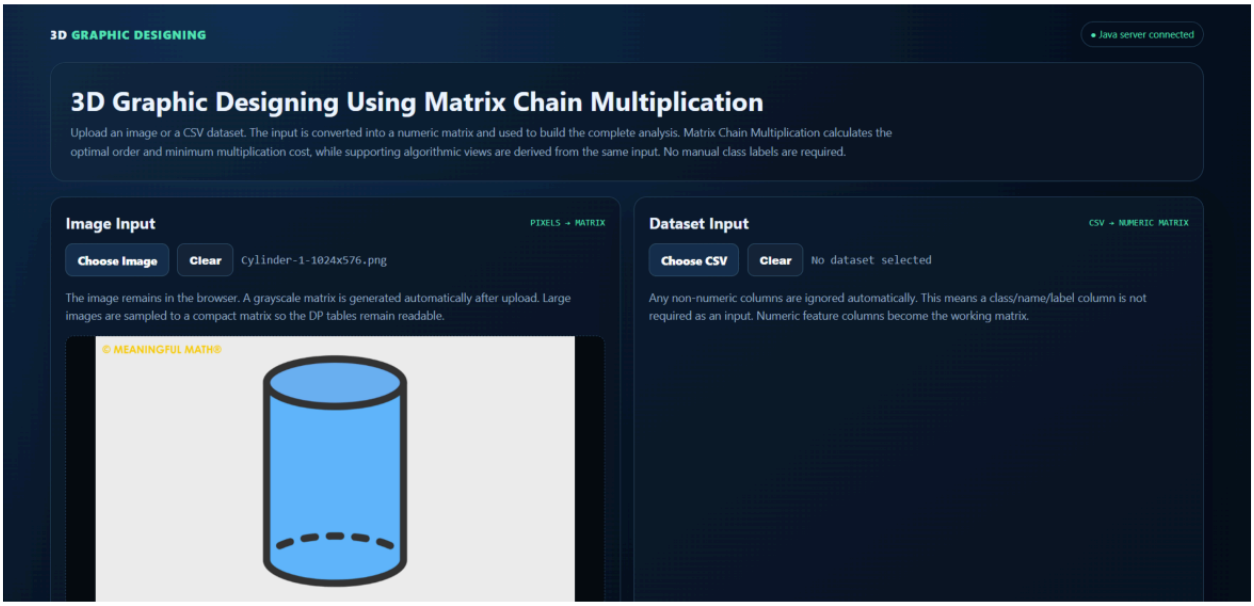
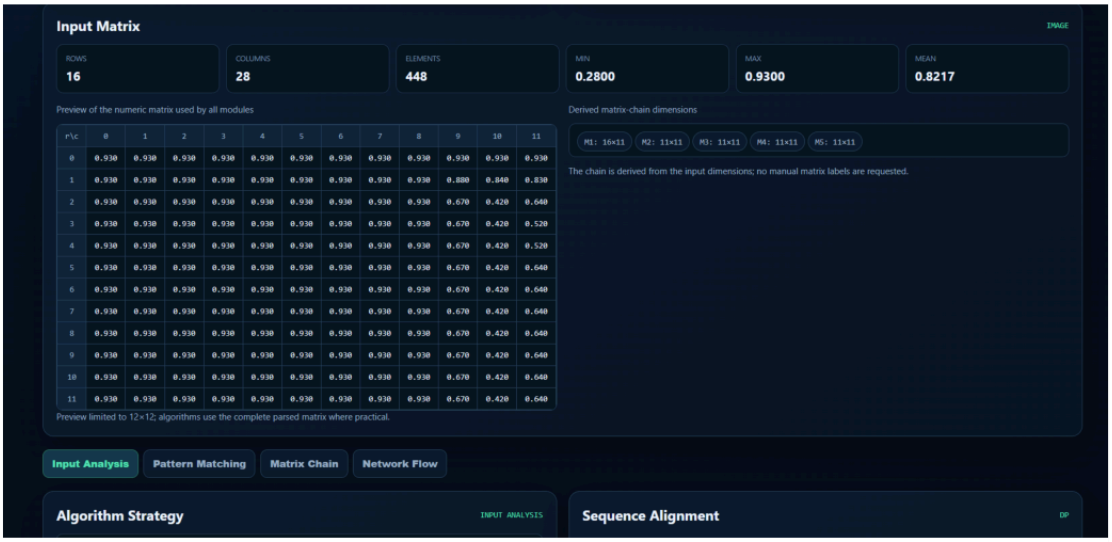
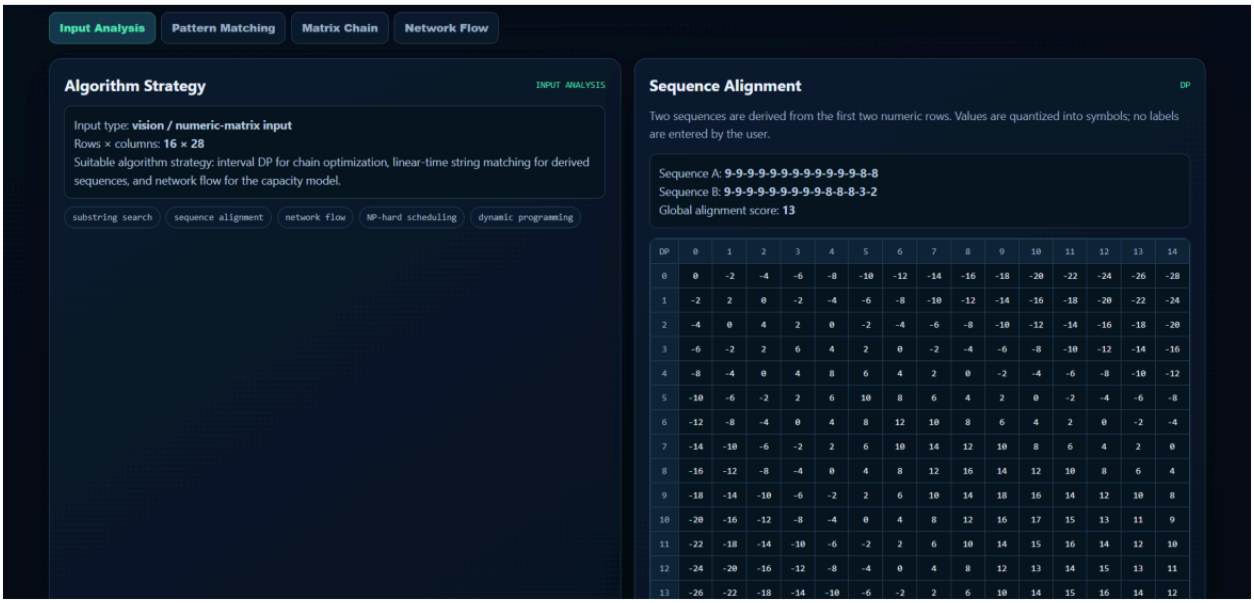
MATRIX CHAIN OPTIMIZER 3D
Interval Dynamic Programming Visualizer

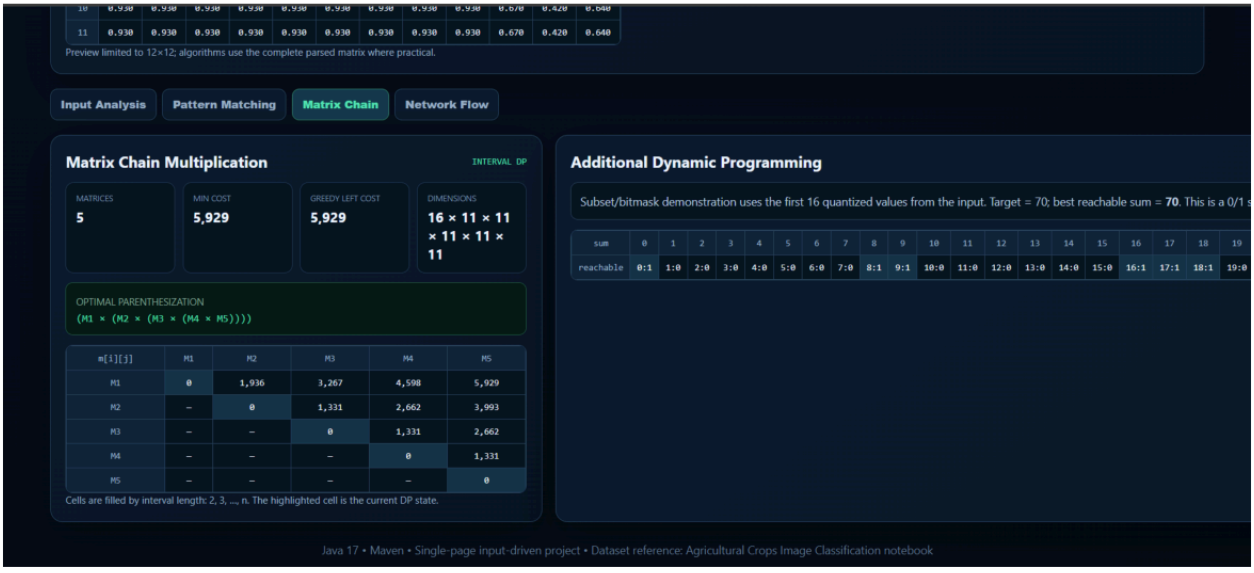
Outputs:







DP	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	0	-2	-4	-6	-8	-10	-12	-14	-16	-18	-20	-22	-24	-26	-28
1	-2	2	0	-2	-4	-6	-8	-10	-12	-14	-16	-18	-20	-22	-24
2	-4	0	4	2	0	-2	-4	-6	-8	-10	-12	-14	-16	-18	-20
3	-6	-2	2	6	4	2	0	-2	-4	-6	-8	-10	-12	-14	-16
4	-8	-4	0	4	8	6	4	2	0	-2	-4	-6	-8	-10	-12
5	-10	-6	-2	2	6	10	8	6	4	2	0	-2	-4	-6	-8
6	-12	-8	-4	0	4	8	12	10	8	6	4	2	0	-2	-4
7	-14	-10	-6	-2	2	6	10	14	12	10	8	6	4	2	0
8	-16	-12	-8	-4	0	4	8	12	16	14	12	10	8	6	4
9	-18	-14	-10	-6	-2	2	6	10	14	18	16	14	12	10	8
10	-20	-16	-12	-8	-4	0	4	8	12	16	17	15	13	11	9
11	-22	-18	-14	-10	-6	-2	2	6	10	14	15	16	14	12	10
12	-24	-20	-16	-12	-8	-4	0	4	8	12	13	14	15	13	11
13	-26	-22	-18	-14	-10	-6	-2	2	6	10	14	15	16	14	12



11	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640

m[i][j]	M1	M2	M3	M4	M5
M1	0	1,936	3,267	4,598	5,929
M2	–	0	1,331	2,662	3,993
M3	–	–	0	1,331	2,662
M4	–	–	–	0	1,331
M5	–	–	–	–	0

sum	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
reachable	0:1	1:0	2:0	3:0	4:0	5:0	6:0	7:0	8:1	9:1	10:0	11:0	12:0	13:0	14:0	15:0	16:1	17:1	18:1	19:0

Input Analysis | Pattern Matching | Matrix Chain | Network Flow

Capacity Network

INPUT-DERIVED

The capacity network is derived directly from the numeric matrix. Source = 0; sink = 7; nodes = 8. Positive directed capacities are created from pairwise input values.

u\v	0	1	2	3	4	5	6	7
0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	1
2	0	0	0	1	1	1	1	1
3	0	0	0	0	1	1	1	1
4	0	0	0	0	0	1	1	1
5	0	0	0	0	0	0	1	1
6	0	0	0	0	0	0	0	1
7	0	0	0	0	0	0	0	0

Edmonds-Karp

MAX-FLOW / MIN-CUT

Maximum flow: **7**
Augmenting paths: **7**
Residual graph computed after all augmentations.

```
Step 1: 0 → 7  |  bottleneck 1  |  total 1
Step 2: 0 → 1 → 7  |  bottleneck 1  |  total 2
Step 3: 0 → 2 → 7  |  bottleneck 1  |  total 3
Step 4: 0 → 3 → 7  |  bottleneck 1  |  total 4
Step 5: 0 → 4 → 7  |  bottleneck 1  |  total 5
Step 6: 0 → 5 → 7  |  bottleneck 1  |  total 6
Step 7: 0 → 6 → 7  |  bottleneck 1  |  total 7
```

Java 17 • Maven • Single-page input-driven project • Dataset reference: Agricultural Crops Image Classification notebook